

High mobility group box-1 protein induces the migration and activation of human dendritic cells and acts as an alarmin

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Abstract: High mobility group box-1 (HMGB1) protein is a nonhistone, DNA-binding protein that plays a critical role in regulating gene transcription. Recently, HMGB1 has also been shown to act as a late mediator of endotoxic shock and to exert a variety of proinflammatory, extracellular activities. Here, we report that HMGB1 simultaneously acts as a chemoattractant and activator of dendritic cells (DCs). HMGB1 induced the migration of monocyte-derived, immature DCs (Mo-iDCs) but not mature DCs. The chemotactic effect of HMGB1 on iDCs was pertussis toxin-inhibitable and also inhibited by antibody against the receptor of advanced glycation end products (RAGE), suggesting that HMGB1 chemoattraction of iDCs is mediated by RAGE in a Gi protein-dependent manner. In addition, HMGB1 treatment of Mo-iDCs up-regulated DC surface markers (CD80, CD83, CD86, and HLA-A,B,C), enhanced DC production of cytokines (IL-6, CXCL8, IL-12p70, and TNF- α), switched DC chemokine responsiveness from CCL5-sensitive to CCL21-sensitive, and acquired the capacity to stimulate allogeneic T cell proliferation. Based on its dual DC-attracting and -activating activities as well as its reported capacity to promote an antigen-specific immune response, we consider HMGB1 to have the properties of an immune alarmin. *J. Leukoc. Biol.* 81: 59–66; 2007.

Key Words: RAGE · chemotaxis · maturation · HMGB1

INTRODUCTION

High mobility group box protein 1 (HMGB1) is a nonhistone, chromatin-binding protein, which was isolated from calf thymus more than 30 years ago [1]. HMGB1, an abundant, 215 amino acid residue-containing nuclear protein present in all mammalian tissues and cells, also known as amphoterin, is highly conserved among various species and structurally organized into two DNA-binding domains (termed A and B box) and a negatively charged C-terminal tail [2, 3]. Studies focusing on its capacity to bind and/or to introduce sharp bends or kinks in DNA double helix and to interact with other transcriptional

factors have established HMGB1 as an important architectural facilitator of the assembly of nucleoprotein complexes that regulate gene recombination and transcription [4–6]. Knocking-out of the mouse *Hmgb1* gene results in the postnatal death of *Hmgb1*^{–/–} mice as a result of a lack of expression of the glucocorticoid receptor [7].

In 1999, HMGB1 was discovered to act as a late mediator of endotoxic shock [8] that has nurtured the subsequent search for the extrachromosomal activities and functions of HMGB1, which can be released by necrotic cells [9, 10] or secreted by activated macrophages [8, 11–13], dendritic cells (DCs) [14], or NK cells [15]. It has been suggested that the secretion of HMGB1 requires sequential acetylation, relocation from nucleus to cytosol, sorting to secretory lysosome, and extracellular disposal [16], a complex process that can be suppressed by cholinergic agonists [13]. HMGB1 regulates neurite outgrowth by binding to the receptor of advanced glycation end products (RAGE) [17, 18]. Extrachromosomal HMGB1 has been shown to mediate inflammation [3, 8, 13], to regulate the migration of monocytes [19], to induce the migration and proliferation of smooth muscle cells and mesoangioblasts [20, 21], to contribute to DC maturation and induction of immune responses [3, 10, 14, 22, 23], and to exert a direct, antibacterial effect [24].

Investigation of the effects of antimicrobial peptides and proteins such as defensins, cathelicidin, and eosinophil-derived neurotoxin on leukocytes have revealed that apart from their direct, antimicrobial activity, these so-called “endogenous antibiotics” also have direct chemotactic and activating activities for various subpopulations of leukocytes including DCs in vitro and can promote an antigen-specific, immune response when coadministered with a given antigen in vivo (reviewed in ref. [25]). Based on their rapid production by cells of the innate immune system in response to infection or tissue injury, their dual roles as chemoattractant and activators for antigen-presenting DCs, as well as their capacity to enhance an antigen-specific immune response, we have proposed to classify those endogenous mediators as alarmins, which rapidly galvanize inflammatory and host defense against exogenous

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danger signals [26–28]. HMGB1 possesses some of the characteristics of alarmins [10, 23]; however, it is not clear whether HMGB1 can induce the chemotaxis and activation of DCs simultaneously. In this study, we found that HMGB1 not only chemoattracted human monocyte-derived DCs (Mo-DCs) in a pertussis toxin (PTX)-sensitive and RAGE-dependent manner but also stimulated the activation of DCs. Thus, we propose to classify HMGB1 as an alarmin.

MATERIALS AND METHODS

Reagents

Human recombinant (r)GM-CSF (specific activity, $\geq 10^7$ U/mg), IL-4 (specific activity, $\geq 2 \times 10^6$ U/mg), stromal cell-derived factor- α /CXCL12, RANTES/CCL5, and secondary lymphoid-organ chemokine/CCL21 were purchased from PeproTech (Rocky Hill, NJ). BSA, fMLP, PTX, FITC-conjugated goat anti-mouse IgG, and LPS (*Escherichia coli*, serotype O55:B5) were from Sigma Chemical Co. (St. Louis, MO). ^3H -TdR (specific radioactivity, 2 Ci/mmol) was purchased from NEN (Boston, MA). Monoclonal antihuman RAGE was purchased from R&D Systems (Minneapolis, MN, MAB11451). RPMI 1640, glutamine, penicillin, streptomycin, and HEPES were from BioWhittaker (Walkersville, MD). FBS was purchased from Hyclone (Logan, UT). All antibodies used for flow cytometry analysis were purchased from BD/PharMingen (San Diego, CA), including FITC-conjugated mouse antihuman CD83 (IgG1, κ , Clone HB15e), FITC-conjugated mouse antihuman CD80 (IgG1, κ , Clone L307.4), FITC-conjugated mouse antihuman HLA-A,B,C (IgG1, κ , Clone G46-2.6), PE-conjugated mouse antihuman CD11c (IgG1, Clone B-Ly6), PE-conjugated mouse antihuman CD86 (IgG1, κ , Clone 2331), PE-conjugated mouse antihuman HLA-DR (IgG2a, κ , Clone G46-6), as well as FITC- and/or PE-conjugated, isotype-matched mouse IgG1, κ , and IgG2a, κ . rHMGB1 was produced as described previously [29]. Native HMGB1 (nHMGB1) was purified from calf thymus exactly, according to the reported method [30]. The endotoxin level in nHMGB1 was measured by the use of *Limulus* amoebocyte lysate (LAL) pyrogen test kit (BioWhittaker) following the manufacturer's protocol. The sensitivity of the test kit for endotoxin was 0.06 IU/ml.

SDS-PAGE and Western blot analysis

SDS-PAGE and Western blot analysis were performed as described previously [26]. Briefly, HMGB1 sample was loaded at 1 $\mu\text{g}/\text{lane}$ onto a 4–12% gradient mini NuPAGE™ Bis-Tris gel (Invitrogen, Carlsbad, CA) using SeeBlue® Plus2 (Invitrogen) as molecular size markers. After electrophoresis, part of the gel was stained with Coomassie Brilliant blue to examine the purity of HMGB1, and the other part of the gel was transferred to a piece of Immobilon™ membrane (Millipore, Bedford, MA) for Western blot. The membrane was blocked, sequentially reacted with rabbit anti-HMGB1 (Abcam, Cambridge, MA) and HRP-conjugated, anti-rabbit IgG (Cell Signaling Technology, Beverly, MA, Cat. #7074), and visualized using ECL Plus detection reagents (Amersham, Piscataway, NJ).

Isolation and purification of cells

Human PBMC were isolated by Ficoll density gradient centrifugation from leukopacks supplied by the Department of Transfusion Medicine (Clinical Center, National Institute of Health, Bethesda, MD) [31]. Monocytes were purified ($>95\%$) from human PBMC with MACS CD14 monocyte isolation kit (Miltenyi Biotec Inc., Auburn, CA). Human peripheral blood T cells were isolated from human PBMC by using the CD3 enrichment column (R&D Systems) following the manufacturer's instruction.

DC culture

DCs were generated as described previously [32]. In brief, monocyte-derived immature DCs (Mo-iDCs) were generated by incubating purified peripheral blood monocytes at $2\text{--}5 \times 10^5/\text{ml}$ in G4 medium (RPMI 1640 containing 10% FBS, 2 mM glutamine, 25 mM HEPES, 100 U/ml penicillin, 100 $\mu\text{g}/\text{ml}$ streptomycin, 50 ng/ml GM-CSF, and 50 ng/ml IL-4) at 37°C in a CO_2 (5%)

incubator for 6–7 days. Subsequently, DCs were incubated in fresh G4 medium in the absence or presence of HMGB1 or LPS at specified concentrations for 24–48 h in a CO_2 (5%) incubator before analysis of their expression of surface markers, production of cytokines in the culture supernatants, and their capacity to stimulate the proliferation of allogeneic T cells.

Chemotaxis assay

DC migration was assessed using a 48-well microchemotaxis chamber as described previously [33]. In brief, DCs were washed three times and resuspended in chemotactic medium (CM; RPMI 1640 containing 1% BSA). HMGB1 and other chemotactic factors were diluted with CM. Different concentrations of chemotactic factors were placed in wells of the lower compartment of the chamber (Neuro Probe, Cabin John, MA), and DC suspension (1×10^6 cells/ml) was added in wells of the upper compartment. In some experiments, HMGB1 or CCL5 was added together with DCs into the upper wells to investigate cross-desensitization. The lower and upper compartments were separated by a 5- μm polycarbonate filter (Osmonics, Livermore, CA). After incubation at 37°C for 1.5 h in humidified air with 5% CO_2 , the filters were removed and stained, and the cells migrated across the filter were counted with the use of a Bioquant semiautomatic counting system. The results were presented as the number of cells per high power field (No./HPF).

ELISA

IL-6, CXCL8, IL-12p70, and TNF- α in the culture supernatants were measured using ELISA kits purchased from R&D Systems, strictly following the protocols provided by the manufacturer.

Flow cytometry

DCs ($10^6/\text{sample}$) were first washed three times with FACS buffer (PBS, 1% FBS, 0.02% NaN_3 , pH 7.4). After blocking at room temperature for 15 min in FACS buffer containing 1% human AB serum and 1% normal mouse serum, DCs were incubated with various FITC- or PE-conjugated mouse mAb against human CD11c, CD80, CD83, CD86, HLA-A,B,C, HLA-DR, or isotype-matched control antibody at room temperature for 30 min as specified. In some experiments, DCs were reacted with anti-RAGE followed by FITC-conjugated goat antimouse IgG before staining with PE-conjugated, antihuman CD83. After washing twice with FACS buffer and twice with PBS, the cells were fixed with 1% paraformaldehyde in PBS and analyzed the next day with a FACScan cytometer (Becton Dickinson, San Jose, CA).

Mixed lymphocyte reaction (MLR)

Allogeneic MLR was performed as described [34]. Briefly, purified, allogeneic T cells ($10^5/\text{well}$) were cultured with different numbers of DCs in a 96-well, flat-bottom plate for 6 days at 37°C in humidified air with 5% CO_2 . The proliferative response of T cells was examined by pulsing the culture with ^3H -TdR (0.5 $\mu\text{Ci}/\text{well}$) for the last 18 h before harvesting. ^3H -TdR incorporation was measured with a microbeta counter (Wallac, Gaithersburg, MD).

Statistical analysis

All experiments were performed two to three times, and the data of one representative experiment are shown. The statistical significance of difference between groups was analyzed using unpaired Student's *t*-test.

RESULTS

The quality of HMGB1

The quality of rHMGB1 was documented previously [29]. The potential endotoxin level in nHMGB1 used in the present study was below the detection limit of the LAL assay, indicating that the nHMGB1 preparation contained less than 0.6 ng endotoxin per mg protein. Analysis of nHMGB1 by SDS-PAGE followed by Coomassie Brilliant blue staining revealed a single band of ~ 30 kDa, suggesting

high purity of the preparation (**Fig. 1**, left panel). When part of the gel was transferred to a PVDF membrane and blotted with anti-HMGB1, a single band with essential identical size was detected, indicating the identity of the band was indeed HMGB1 (**Fig. 1**, right panel).

HMGB1 is a chemoattractant for Mo-DCs

To test if HMGB1 could act as a chemoattractant for DCs, we first examined if HMGB1 could induce the migration of Mo-iDCs using a commonly used, 48-well microchemotaxis chamber assay with CXCL12 as a positive control (**Fig. 2**). nHMGB1 induced the migration of Mo-iDCs in a dose-dependent manner with a peak response at 10 ng/ml (**Fig. 2A**). rHMGB1 showed similar activity (**Fig. 2B**). Even at the optimal concentration of 10 ng/ml, the number of iDCs migrated in response to nHMGB1 or rHMGB1 was approximately half of that migrated in response to an optimal concentration of CXCL12 (**Fig. 2A**) or CCL5 (**Fig. 2B**), suggesting that HMGB1 was less efficacious than CXCL12 and CCL5 in terms of attracting iDCs. Neither nHMGB1 nor rHMGB1 at the concentrations tested induced the migration of mature DCs (mDCs), which as expected, did not migrate in response to CCL5 but migrated in response to CCL21 (**Fig. 2C**).

To test whether HMGB1-induced migration of Mo-iDCs was a result of chemotaxis or chemokinesis, we determined if addition of HMGB1 together with iDCs into the upper

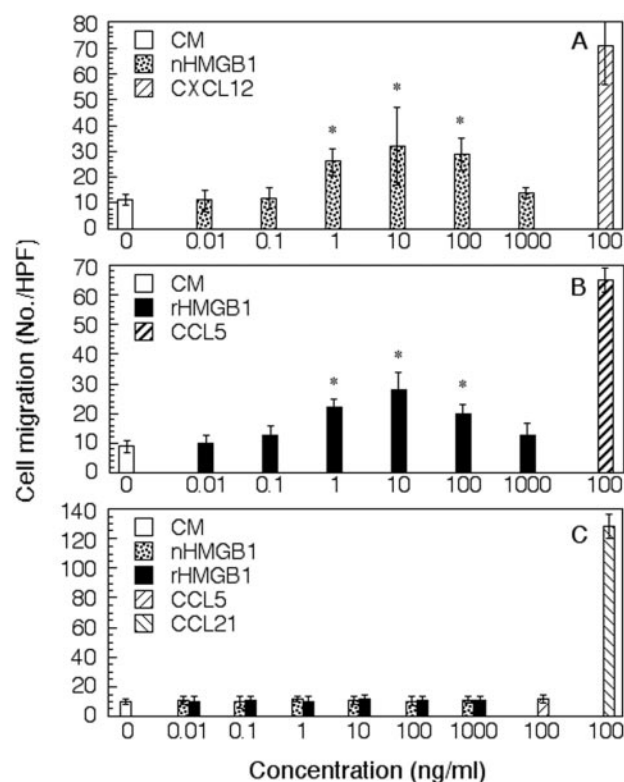


Fig. 2. HMGB1 induction of DC migration. Human Mo-iDC migration, in response to nHMGB1 (A) or rHMGB1 (B). Shown are the results (mean $\pm$ SD of triplicate wells) of one experiment representative of three to four performed with iDCs generated from monocytes isolated from different healthy donors. *, $P < 0.05$, when compared with background migration (open bar). (C) Migration of DCs matured by treatment with LPS (1 μ g/ml) for 48 h in response to nHMGB1, rHMGB1, and selected chemokines. CXCL12, CCL5, and CCL21 were used at their optimal concentrations (100 ng/ml).

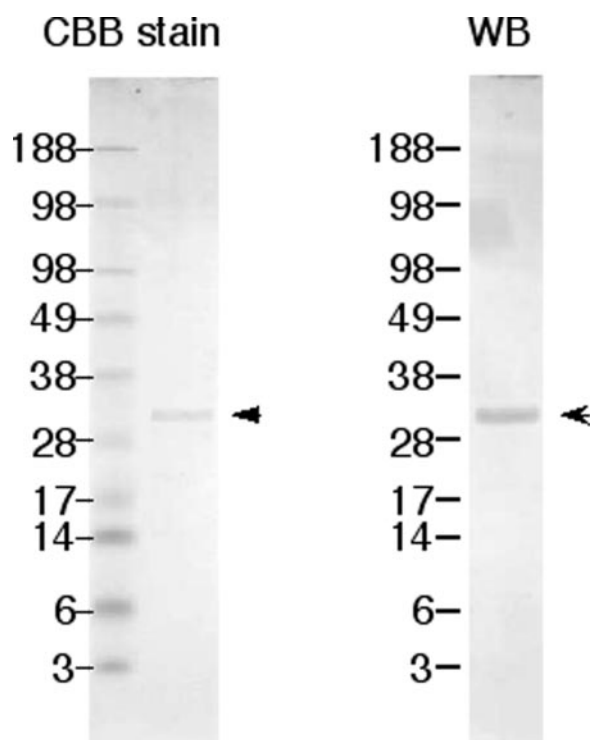


Fig. 1. SDS-PAGE and Western blot analysis of nHMGB1 preparation. nHMGB1 sample (1 μ g) was loaded into a gradient mini-SDS-PAGE gel with SeeBlue2 as a molecular size marker. After electrophoresis, the gel was sliced, one lane was stained with Coomassie Brilliant blue (CBB; left panel), and one lane was Western blotted (WB) with anti-HMGB1 after transfer onto a piece of polyvinylidene difluoride (PVDF) membrane (right panel). The bands are marked by arrowheads.

wells of the chemotaxis chamber could desensitize the migration of iDCs in response to HMGB1 in the lower wells. As shown in **Figure 3**, compared with background migration of iDCs (first bar), nHMGB1 and CCL5 in the lower wells induced migration of iDCs (second and fifth bars, respectively). Addition of identical concentration of nHMGB1 into the upper wells significantly suppressed the migration of iDCs in response to nHMGB1 in the lower wells, suggesting homologous desensitization (fourth bar). The desensitization was not a result of inhibition of cell motility, as identical concentration of nHMGB1 in the upper wells did not reduce the background migration (third bar). In addition, the desensitization was specific, as neither nHMGB1 desensitized CCL5-induced iDC migration (sixth bar) nor did CCL5 desensitize nHMGB1-induced iDC migration (seventh bar). Similar results were obtained with the use of rHMGB1 (data not shown). These results established HMGB1 as a chemoattractant for iDCs.

HMGB1 chemoattracts iDCs in a Gi-protein- and RAGE-dependent manner

Most chemokines and chemoattractants, upon binding to their receptors, cause the activation of heterotrimeric Gi protein,

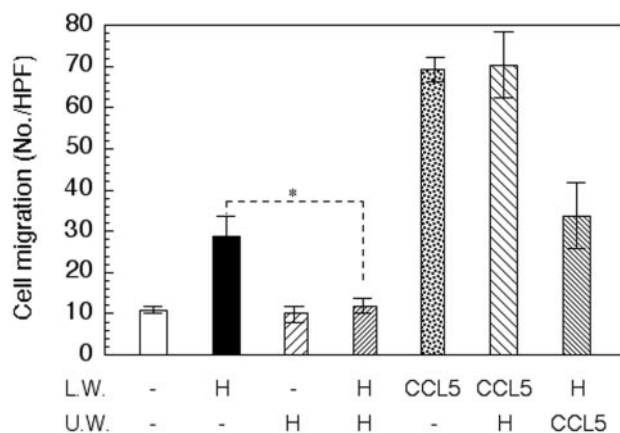


Fig. 3. Homologous desensitization of HMGB1-induced DC migration. The migration of Mo-iDCs in response to nHMGB1 in the lower wells was inhibited by addition of an identical amount of nHMGB1 into the upper wells of a chemotaxis chamber. Shown is the average (mean $\pm$ SD) cell migration (No./HPF) of three wells per group. L.W., Lower well; U.W., upper well; H, nHMGB1. The concentrations used for nHMGB1 and CCL5 were 10 and 100 ng/ml, respectively. Similar results were obtained from three separate experiments. *, $P < 0.05$.

which leads to chemoattraction of leukocytes [35]. To determine if HMGB1-induced iDC chemotaxis was mediated by a Gi protein, we examined whether HMGB1-induced iDC chemotaxis could be inhibited by PTX, a toxin that specifically inhibits the activation of the Gi protein by adenosine 5'-diphosphate-ribosylating the $G_{i\alpha}$ subunit of the heterotrimeric G protein [36]. Pretreatment of iDCs with PTX at 200 ng/ml prior to chemotaxis assay inhibited the migration of iDCs in response to nHMGB1 (**Fig. 4A**) and rHMGB1 (data not shown). Pretreatment with PTX did not affect the motility of monocytes, as there was no difference in spontaneous migration between PTX-pretreated and -untreated iDCs. As expected, CCL5-induced iDC migration was also inhibited by PTX.

The migration of muscle cells in response to HMGB1 is mediated by RAGE [20, 21]. To test if HMGB1-induced iDC migration was also dependent on RAGE, we examined if anti-RAGE antibody could suppress iDC chemotaxis in response to HMGB1 (**Fig. 4B**). Although isotype-matched control mouse IgG2b did not, anti-RAGE antibody dose-dependently reduced nHMGB1-induced migration of iDCs with significant inhibition at 5 and 20 μ g/ml, suggesting the participation of RAGE in mediating the chemotactic effect of HMGB1 on DCs. As HMGB1 induced the migration of iDCs but not mDCs (**Fig. 2**), we further investigated whether the failure of mDCs to migrate to HMGB1 might be a result of a down-regulation of RAGE expression on the surface of mDCs by flow cytometry. As expected, iDCs were stained negative for CD83 and positive for RAGE (**Fig. 4C**, upper panels). DCs treated with LPS not only became mature ($CD83^+$) but also significantly reduced their surface expression of RAGE (**Fig. 4C**, lower panels). Therefore, HMGB1-induced DC chemotaxis seemed to be dependent on Gi protein and RAGE.

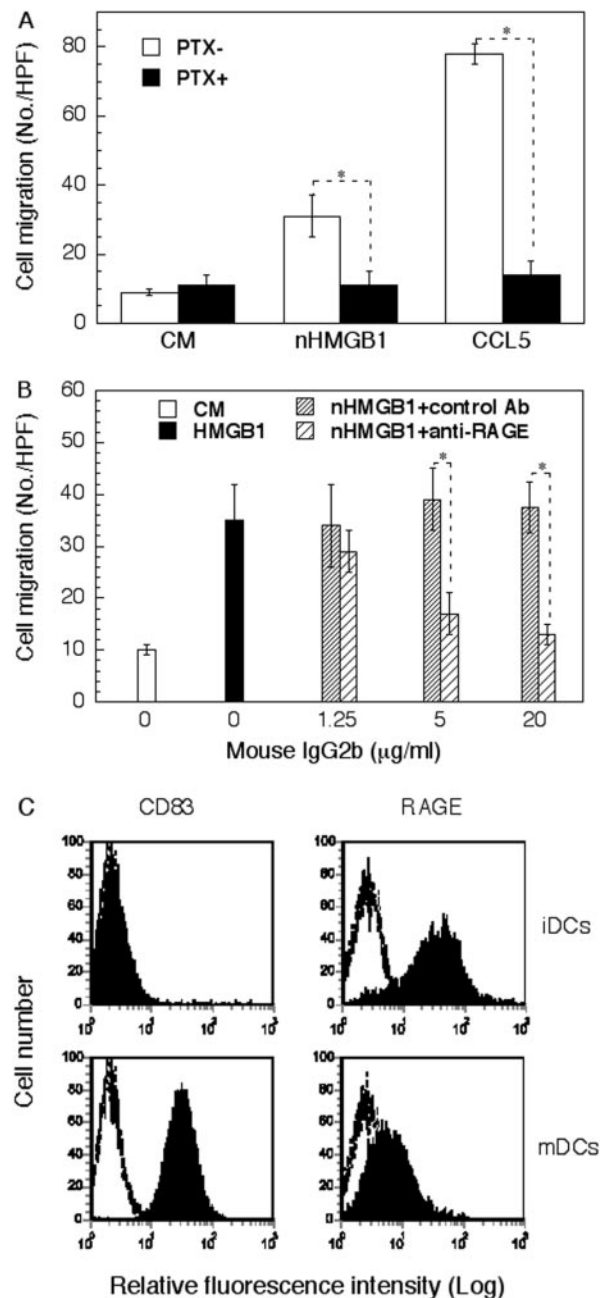


Fig. 4. Dependence of HMGB1-induced iDC chemotaxis on Gi protein and RAGE. (A) Inhibition of HMGB1-induced iDC migration by PTX. Mo-iDCs at 10^6 /ml in CM were incubated in the absence or presence of PTX at a final concentration of 200 ng/ml for 30 min before they were used for the chemotaxis assay. nHMGB1 and CCL5 were used at the concentration of 50 and 100 ng/ml, respectively. *, $P < 0.05$. (B) Suppression of HMGB1-induced chemotaxis of Mo-iDCs by anti-RAGE, and anti-RAGE or isotype-matched control antibody was added together with iDCs into the upper wells of the chemotaxis chamber at concentrations as specified. The concentration of nHMGB1 added into the lower wells was 10 ng/ml. *, $P < 0.05$. (C) Surface expression of RAGE by DCs. Mo-DCs were incubated in the absence or presence of LPS (1 μ g/ml) for 48 h to generate iDCs and mDCs, respectively. Subsequently, iDCs and mDCs were stained with anti-CD83 and anti-RAGE and analyzed by flow cytometry as described in Materials and Methods. Open and solid histograms illustrate the staining with isotype-matched control and specific antibody, respectively.

HMGB1 induces phenotypic and functional activation of DCs

To investigate whether nHMGB1 could induce DC activation simultaneously, Mo-DCs were treated with nHMGB1 at specified concentrations, and the resulting DCs were analyzed for phenotypic characteristics of activated DCs. Fully activated DCs acquire a mature phenotype, characterized by the up-regulation of certain surface molecules, switch of chemokine responsiveness, and production of a variety of inflammatory cytokines and chemokines [37]. As shown by **Figure 5A**, DCs treated with nHMGB1 for 48 h, in comparison with sham-treated DCs, became positive for CD83, a marker for DC maturation, and up-regulated the expression of surface molecules essential for antigen presentation, such as CD80, CD86, and HLA-A,B,C. LPS, as a positive control, as expected, also stimulated up-regulation of CD83, CD80, CD86, and HLA-A,B,C expression on the surface of DCs (Fig. 5A). When the capacity of DCs to migrate in response to CCL5, CCL21, and CXCL12 was measured (Fig. 5B), sham-treated DCs migrated to CCL5, but not CCL21, indicative of their immature phenotype (open bars). DCs treated with LPS, a well-known substance capable of inducing DC maturation, migrated to CCL21

but not CCL5 (Fig. 5B, hatched bars). DCs treated with nHMGB1 similarly acquired the capacity to migrate to CCL21 and simultaneously lost the responsiveness to CCL5, suggesting that DCs switched from CCL5-responsive to CCL21-responsive when they underwent HMGB1-induced phenotypic maturation (Fig. 5B, solid bars). CXCL12 was used as a positive control, as it can induce the migration of iDCs and mDCs.

When the supernatants of DCs cultured in the absence or presence of nHMGB1 or LPS were measured for cytokines indicative of DC maturation, nHMGB1 was found to dose-dependently enhance the production of IL-6, CXCL8, IL-12p70, and TNF- α (Fig. 5C). The magnitude of enhancement varied from 100 (CXCL8, IL-12p70, and TNF- α)- to 1000-fold (IL-6). HMGB1-treated DCs produced significantly higher CXCL8, IL-12p70, and TNF- α than DCs treated with 1 μ g/ml LPS (Fig. 5C). Based on its capacity to up-regulate DC surface CD80, CD83, CD86, and HLA-A,B,C, to convert DCs from CCL5-responsive to CCL21-responsive, as well as to promote DC production of proinflammatory cytokines, particularly IL-12p70, it can be concluded that HMGB1 is capable of inducing the phenotypic maturation of DCs.

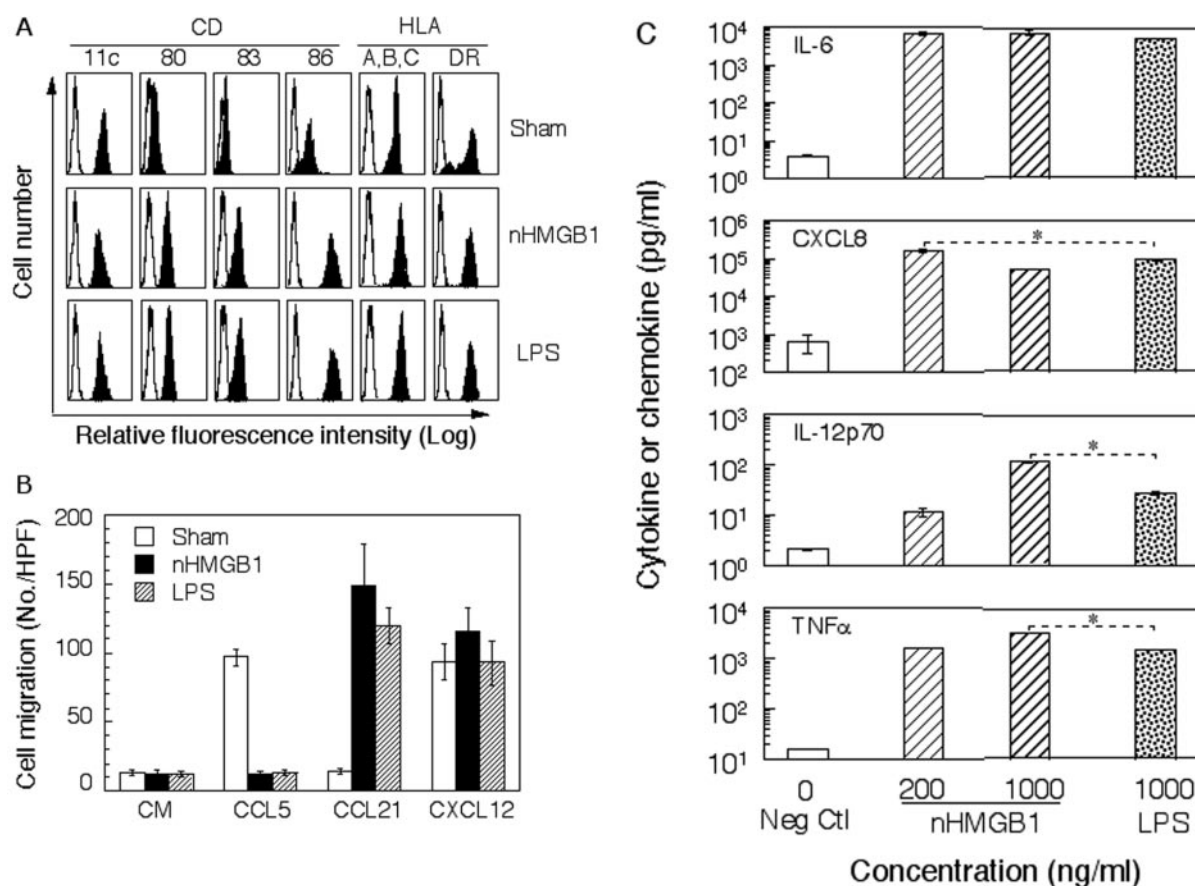


Fig. 5. Induction of phenotypic maturation of DCs by nHMGB1. (A) Flow cytometric analysis of the expression of DC surface markers. Mo-iDCs were incubated in the absence (Sham) or presence of nHMGB1 (2 μ g/ml) or LPS (1 μ g/ml) for 2 days before immunostaining. Open and solid histograms illustrate the staining with isotype-matched control and specific antibody, respectively. (B) Migration of differently treated DCs in response to selected chemokines. Mo-DCs were cultured in the absence (Sham) or presence of nHMGB1 (1 μ g/ml) or LPS (1 μ g/ml) for 24 h before the measurement of their migration to indicated chemokines. CCL5, CCL21, and CXCL12 were used at 100 ng/ml. (C) DC production of cytokines in response to nHMGB1. Mo-iDCs (5×10^5 cells/ml) were incubated in triplicate in the absence or presence of HMGB1 and LPS at the concentrations specified for 48 h before the supernatant was harvested for the measurement of IL-6, CXCL8, IL-12p70, and TNF- α by ELISA. Shown are the results (mean $\pm$ SD) of one experiment representative of two. *, $P < 0.05$.

To make certain whether HMGB1-induced maturation of DCs was reflected at the functional level, nHMGB1-treated DCs were analyzed for their capacity to stimulate the proliferation of T cells in an allogeneic, MLR setting using sham-treated DCs as a control (**Fig. 6**). At the DC:T ratio tested, sham-treated DCs did not stimulate the proliferation of allogeneic T cells, as judged by no significant increase in ^3H -TdR incorporation. DCs treated with nHMGB1, however, began to stimulate the proliferation of allogeneic T cells when used at a DC:T ratio higher than 1:6250, suggesting that HMGB1-treated DCs exhibited a remarkably enhanced capacity for presenting antigen to T cells for the stimulation of cell proliferation and activation. Thus, HMGB1 treatment results in full activation of DCs, which not only acquire a mature phenotype (**Fig. 5**) but also become capable of antigen presentation and T cell stimulation (**Fig. 6**).

DISCUSSION

One of the major observations in this study is the identification of HMGB1 as a chemoattractant selective for iDCs but not mDCs. This is based on the bell-shaped, dose-response curve of HMGB1-induced migration of iDCs (**Fig. 2**) and selective inhibition of HMGB1-induced iDC migration by addition of an identical concentration of HMGB1 into the upper wells of the chemotaxis assay (**Fig. 3**). Previously, Bianchi's group [20, 21] has shown that HMGB1 is chemotactic for smooth muscle cells and mesoangioblasts. In their studies, the migration of smooth muscle cells in response to HMGB1 was suppressed by anti-RAGE antibody, dominant-negative RAGE, as well as PTX [20, 21]. Our data showing that HMGB1-induced migration of iDCs was also inhibited by PTX and anti-RAGE antibody (**Fig. 4**) indicate that HMGB1 induction of iDC migration is also mediated by RAGE in a Gi protein-dependent manner. The failure of mDCs to migrate in response to HMGB1 can be the consequence of several reasons. One possibility is the down-

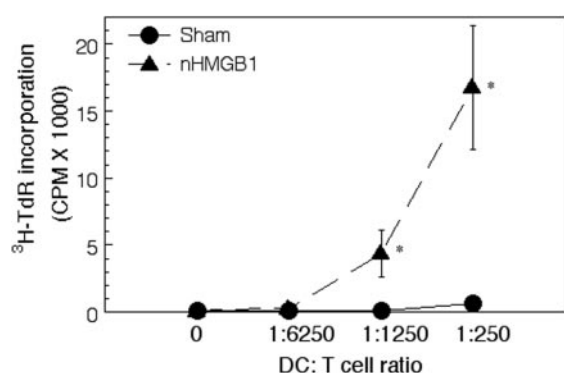


Fig. 6. Stimulation of the proliferation of allogeneic, human peripheral blood T lymphocytes by HMGB1-treated DCs. T cells (10^5 /well) were cultured in triplicate in the absence or presence of DCs at concentrations specified in 96-well plates for 6 days with the addition of ^3H -TdR (0.5 μCi /well) in the last 18 h of incubation. The cells were harvested and measured for the incorporation of ^3H -TdR (mean $\pm$ SD). DCs were treated with or without (sham) 1 $\mu\text{g}/\text{ml}$ nHMGB1 for 48 h at 37°C in humidified air with 5% CO_2 for 48 h before used in the allogeneic MLR experiments. *, $P < 0.01$, when compared with the corresponding sham group.

regulation of the receptor(s) that mediates the chemotactic effect of HMGB1, as DCs matured by LPS, in comparison with iDCs, exhibited a decrease in RAGE expression on the surface (**Fig. 4C**).

Another major observation of this study is that nHMGB1 has the capacity to stimulate the activation of Mo-DCs. Activated (or fully matured) DCs, in comparison with iDCs, demonstrate a mature phenotype and the capacity to stimulate the proliferation and activation of T cells [25, 37]. rHMGB1 was shown previously to stimulate the maturation of DCs [23]. The data showing that DCs, upon treatment with nHMGB1, up-regulate the expression of costimulatory molecules (CD80 and CD86), HLA-A,B,C, and the maturation marker CD83 enhance the production of proinflammatory cytokines including IL-12 and a shift from CCL5-responsive to CCL21-responsive and acquire the capacity to stimulate the proliferation of T cells, indicating that nHMGB1 is also capable of stimulating the maturation of activation of DCs (**Figs. 5 and 6**).

How are the dual DC-attracting and -activating effects of HMGB1 mediated? As HMGB1 induction of neurite outgrowth [17, 18], monocyte movement [19], the migration of smooth muscle cells [20, 21], and DCs (**Fig. 4**) can be inhibited by blockade of RAGE, RAGE appears to be the receptor for mediating the effect of HMGB1 on the migration of various cells including DCs. However, HMGB1-induced migration of DCs (**Fig. 4**) and smooth muscle cells [20] can be inhibited by PTX, suggesting a Gi protein is also involved in HMGB1-induced cell migration. As Gi proteins are known to mediate the signaling of seven-transmembrane domain receptors, such as those for chemokines and classic chemoattractants [35, 36], and PTX is not known to inhibit RAGE-mediated signaling, it cannot be ruled out that HMGB1 may induce cell migration by interacting with an unidentified Gi protein-coupled seven-transmembrane domain receptor, while its binding to RAGE helps presenting HMGB1 to its receptor.

In addition to stimulating DC activation (the present study and refs. [10, 14, 22, 23]), HMGB1 activates macrophage and neutrophils [8, 38–40]. The capacity of HMGB1 to activate macrophages and to mediate septic shock can be suppressed by blockade of RAGE [39, 41, 42], suggesting that RAGE also participate in the cell-activating effect of HMGB1. Nevertheless, the suppression of HMGB1-induced macrophage activation by anti-RAGE antibodies is often incomplete. Other receptor(s), such as TLR2 and TLR4, are also suggested to mediate HMGB1-induced macrophage activation and intracellular signaling [38]. Most recently, using fluorescence resonance energy transfer, coimmunoprecipitation, and human embryonic kidney (HEK)293 cells transfected to express TLR2, TLR4, or RAGE, Park et al. [43] have shown physical interaction between HMGB1 and TLR2 and -4 (but not RAGE) on the cell membrane of macrophages and transfected HEK293 cells. Moreover, they have also demonstrated the dependence of HMGB1-induced NF- κB activation in transfected HEK293 cells on TLR2 and -4 rather than RAGE [43]. Thus, the basis for HMGB1-induced activation of cells including DCs may involve the use of multiple receptors, including RAGE and TLRs.

Many endogenous mediators such as defensins, cathelicidins, eosinophil-associated RNases, granulysin, and certain

chemokines, although structurally distinct and exhibiting diverse activities toward multiple targets, share the following characteristics: They are released rapidly in response to microbial encounter and/or tissue injury; they exhibit dual-capacity of mobilizing and activating APC, in particular, DCs; and they are capable of enhancing the antigen-specific, adaptive-immune response in vivo. These endogenous mediators have been proposed to behave as immune alarmins [26–28]. HMGB1 has antibacterial activity [24], is released rapidly upon necrotic cell death [9], and is capable of enhancing antigen-specific immune response [10]. The identification of the DC-chemoattracting activity of HMGB1 in the present study suggests that it is capable of recruiting or mobilizing DCs to its site of release. Furthermore, HMGB1 is capable of activating APC (the present study and refs. [8, 10, 15, 23, 39]). Thus, HMGB1 can also be considered an alarmin.

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